

Smart Farmer Assistance System

Step-by-Step Practical User Guide & Operations Manual

System Overview: The Smart Farmer Assistance System (SFAS) is an AI-powered, offline-capable Progressive Web Application built specifically for Indian farmers. It unifies personalized crop advisory, edge AI disease diagnosis, real-time APMC mandi net profit calculation, localized spray weather forecasts, and government scheme eligibility into an intuitive interface.

1. Getting Started & Sign In

Accessing the portal takes just seconds on both smartphone and computer browsers:

- **Website Access:** Open your deployed Vercel link or local address in any browser (Google Chrome, Brave, Safari, Edge).
- **Language Selection:** Tap the Globe icon in the top navigation bar to choose your preferred language: **English**, **हिंदी (Hindi)**, or **मराठी (Marathi)**. All navigation and terminology updates immediately.
- **Instant Demo Login:** Tap the 'Sign In' button on the top right. You can immediately choose one of the pre-loaded authentic demo profiles without typing passwords:

Farmer Name	District & State	Land & Soil Type	Primary Crop
Ramesh Patil	Nashik, Maharashtra	2.2 Acres, Black Soil (Regur)	Tomato (Flowering)
Rajesh Sharma	Karnal, Haryana	9.5 Acres, Alluvial Loam	Wheat (Vegetative)

2. Navigating the Unified Dashboard

The central dashboard aggregates real-time agricultural telemetry tailored to your farm:

- **Live Weather Card:** Displays current temperature, humidity, wind velocity, and an automated spray advisory status.
- **My Farm Status:** Displays registered acreage, active crop, growth stage, and soil health conditions.
- **Urgent Advisory Alerts:** Color-coded banners warning of sudden humidity spikes, pest incubation windows, or mandi price surges.
- **Quick Launch Actions:** One-tap navigation cards to jump directly into Disease Diagnosis, Crop Advisory, Mandi Comparison, and Schemes.

3. Crop Advisor & Net Profit (ROI) Calculator

This module eliminates guesswork when choosing which crop to sow for the upcoming season:

Step 1: Input Your Farm Parameters — Select your Land Area (in Acres), Soil Type (e.g., Black Cotton, Alluvial Loam, Sandy Loam), Water Source, and Season (Kharif, Rabi, Zaid).

Step 2: Generate Recommendations — The system calculates agronomic suitability scores and ranks crops based on water risk, market volatility, and input costs.

Step 3: Analyze Detailed Economics — Click on any recommended crop card (Tomato, Onion, Cotton, Wheat, Green Chilli, Maize, Soybean) to view an interactive financial breakdown:

Economic Metric	Formula / Derivation	Farmer Utility
Gross Expected Revenue	Expected Yield (Quintals) × Mandi Price	Forecasts maximum earning potential
Estimated Input Cost	Seed + Fertilizer + Crop Protection + Labor	Prevents over-leveraging with seed loans
Net Projected Profit	Gross Revenue - Total Input Cost	Highlights true bottom-line earnings
ROI Percentage	(Net Profit ÷ Total Input Cost) × 100	Enables direct crop-to-crop comparison

4. Plant Disease Diagnosis (Edge AI Scanner)

Farmers can detect leaf pathologies in under 3 seconds using the integrated vision scanner:

Step 1: Upload or Capture Photo: Click on '**Upload Leaf Image**' or drag & drop a photo from your gallery. You can use the ready-made test images from your sample-dataset/ folder:

- sample-dataset/tomato_early_blight.jpg (Early Blight with concentric ring lesions)
- sample-dataset/potato_late_blight.jpg (Late Blight with dark water-soaked blotches)
- sample-dataset/healthy_crop_leaf.jpg (Clean, healthy foliage verification)

Step 2: Instant Edge AI Diagnosis: The scanner processes visual features directly in your browser with zero latency. It outputs:

- **Disease Name & Pathogen:** E.g., *Alternaria solani* or *Phytophthora infestans*.
- **Confidence Rating:** E.g., 94% Confidence.
- **Symptom Summary:** Visual pathology description to cross-verify in the field.

Step 3: Actionable Treatment Plan: The report gives both **Organic Remedies** (Neem oil 5ml/L, Trichoderma harzianum) and **Chemical Controls** (Mancozeb, Metalaxyl) with exact dosage instructions.

Step 4: Voice Readout: Tap the **Audio Speaker button** to hear the diagnosis read aloud in Marathi, Hindi, or English.

5. Smart Mandi Prices & Transport Logistics

Selling at the nearest local mandi is often less profitable than selling at a regional hub due to price disparities. This module calculates the true profit after transportation:

- **Live APMC Comparison:** Displays modal prices, minimums, maximums, and daily arrival tonnage across mandis (e.g., Nashik APMC, Lasalgaon Onion Yard, Pune Gultekdi, Vashi Mumbai Terminal).
- **Net Realized Profit Formula:** Automatically deducts (Distance in km × Freight Rate per km) + Mandi Handling & Loading Fee from the gross price.
- **Shared Transport Pooling:** Farmers can join or create shared transport pools with neighboring growers, slashing freight expenses by 35% to 50% for smaller vehicle loads.

6. Weather Forecasts & Spray Advisory Windows

Farming decisions hinge on weather timing. This screen gives agro-meteorological advisories:

- **7-Day Hyper-Local Forecast:** Temperature range, relative humidity percentage, wind speed, and rain probability.
- **Optimal Spray Window Indicator:** A clear Green/Yellow/Red badge tells you whether it is safe to spray fungicides/pesticides today. If rain is expected within 4 hours or wind speed exceeds 18 km/h, the system flags a '*High Washout Risk*' warning to save pesticide costs.
- **Evapotranspiration Guidance:** Alerts you when high heat and dry winds necessitate additional drip irrigation cycles.

7. 24/7 AI Agronomist Assistant

An intelligent digital agronomist trained on university package-of-practices and agricultural ICAR guidelines:

- **Voice & Text Interaction:** Speak or type queries directly in Marathi, Hindi, or English (e.g., '*How to treat yellow leaves in tomato?*' or '■■■■■ ■■■ ■■ ■■■■ ■■ ■■■ ■■■■ ■■■■ ■■■■?').
- **Context-Aware Answers:** Uses your logged-in farm profile (soil type, crop stage) to give tailored, precise recommendations.
- **Audio Response:** Integrated Web Speech synthesizes audio answers so farmers can listen hands-free in the field.

8. Government Schemes & Subsidy Navigator

Finds verified state and central agricultural schemes you are eligible for:

- **Filter by Category:** Direct Income Support, Crop Insurance, Drip Irrigation Subsidies, Farm Mechanization, and Kisan Credit Cards (KCC).
- **Eligibility Breakdown:** View exact land ceiling criteria, required documents (Aadhaar, 7/12 land record, bank passbook), and direct official portal links (e.g., pmkisan.gov.in, pmfby.gov.in).

9. Offline Capabilities & Mobile Installation

Rural connectivity is often intermittent. SFAS is engineered to remain fully operational offline:

- **Automatic Offline Caching:** Once loaded, crop data, mandi prices, farm profiles, and edge AI disease detection run completely without internet.
- **Offline Status Indicator:** An amber badge alerts you whenever the connection drops, assuring you that cached calculations remain active.
- **Install as Mobile App (PWA):**
 1. Open the website on your smartphone Chrome browser.
 2. Tap the three dots menu (⋮) -> Select 'Install App' or 'Add to Home Screen'.
 3. The SFAS app icon appears on your phone launcher, opening in full-screen standalone mode without URL bars.

10. Troubleshooting & Support Checklist

Issue Observed	Probable Cause	Quick Resolution
Prices or Crops not loading	Backend server sleeping or offline	Wait 30s for Render free tier or check connection
Disease scan shows low confidence	Blurry image or bad lighting	Use close-up daylight photos from sample-dataset/
Audio readout not speaking	Browser voice permission muted	Unmute device media volume & grant audio permissions
Offline badge stays visible	No network connection	App uses cached data; it auto-syncs when online

Smart Farmer Assistance System (SFAS) — Empowering farmers through AI, transparency, and data-driven agronomy.